

ИНФРАСТРУКТУРА ИИ / СЕТЕВЫЕ ТЕХНОЛОГИИ / РАЗРАБОТКА ПЛАТФОРМ

От системы учета к системе управления: как NetBox Labs превращает сетевых инженеров в «мастеров намерений».

NetBox Labs помогает сетевым инженерам управлять намерениями, переходя от системы учета к системе контроля, чтобы справляться с дрейфом сети и масштабированием ИИ.

28 апреля 2026 года, 11:00 [Дуг Силларс](#)



Masantocreative для Unsplash+

[NetBox Labs](#) спонсировала этот пост.

Первые интернет-сети были настолько малы, что один инженер мог держать лове всю систему. Вам не нужна была система учета — вы и были такой

провайдерами и сервисами. Текущее состояние сети стало одновременно и невозможным для понимания, и критически важным для обеспечения.

Сетевая инфраструктура стала основой бизнеса, а ошибки в конфигурации — это уже не просто досадные мелочи, а критически важные для бизнеса события. Агенты с искусственным интеллектом еще больше нагружают сетевую инфраструктуру. Как же сетевым инженерам обеспечить ее бесперебойную работу?

Индивидуальное отслеживание не масштабируется

Таблицы Excel и облачные информационные панели уже не справляются. Для работы современной инфраструктуры требуется надежное представление о том, какой *должна быть* сеть — какие системы и сервисы существуют, где они находятся и как они связаны. Без этого управление сетью становится рискованным.

Раньше ваша карта могла уместиться в таблице Excel, а все настройки можно было выполнить в терминале. Но современным сетям нужен не просто мастер за терминалом, а архитекторы, которые могут видеть общую картину и понимать, как взаимодействуют части сети.

«Раньше ваша карта могла уместиться в таблице Excel... Но современным сетям нужен не просто мастер за терминалом, а архитекторы, способные видеть общую картину».

NetBox Labs предоставляет сетевым архитекторам инструменты, необходимые для создания сложных архитектур, в которых моделируется все — от IP-адресов до физических кабелей. Эта схема становится «**системой учета**», на основе которой принимаются решения по дальнейшему развитию сети и формируются системы управления.

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But a system of record alone isn't enough. Knowing what *should* exist doesn't guarantee that's what actually exists.

As modern **networks and infrastructure** are constantly being updated and reconfigured, that change is increasingly driven by intent, modeled in the system of record. NetBox Labs is central to change management, and teams drive configuration changes directly from the platform, ensuring the infrastructure is aligned with the intended design.

TRENDING STORIES

1. [HTTP/3 Is Now a Standard: Why Use It and How to Get Started](#)
2. [Linux: Display and Manage IP Address Settings](#)
3. [How To Read a Traceroute for Network Troubleshooting](#)
4. [From system of record to system of control: How NetBox Labs is making network engineers "masters of intent."](#)
5. [How Bolt Scaled Remote Work with Tailscale's Zero Trust Mesh VPN](#)

But not all changes in the network happen intentionally. How can network engineers surface every change and determine whether it is a meaningful update or drift — changes that deviate from the intended structure?

Are network changes updates or drift?

Network changes can be bucketed (loosely) into two categories: updates (a new service launch) or drift (an engineer pokes open a port for testing and forgets to close it, a team fires up a farm of test servers, and forgets to shut them down)

Updates are great — the network is growing, becoming stronger and more secure. These are planned and generally driven by changes that first occur directly in the system of record, which then drive downstream configuration in the network itself. This ensures the network's truth evolves directly in tandem with live changes.

running the network. Changes that introduce drift must be resolved to bring the network **back into compliance**.

So how do you tell? First, you must have a system of record (provided by NetBox) to know what should be there, and a way to quickly surface changes so your network engineers can identify them as growth or drift.

“What modern networks require is a system of control — a continuous loop that compares intended state to operational reality and acts on the difference.”

What modern networks require is a system of control — a continuous loop that compares intended state to operational reality and acts on the difference.

Reconciling changes

NetBox Discovery’s continuous mapping of the network provides the truth of what is live in the network today. **NetBox Assurance** compares the network’s live state to your system of record.

Assurance automatically discovers changes in the network that deviate from the system of record. These changes are surfaced to network engineers, who are given the opportunity to either update the source of record (this is a planned change to the network) or remediate any changes that have drifted from the system of record. Companies like **CoreWeave** use NetBox Labs to accelerate data center deployments by helping them maintain their source of truth.

The future of network engineering

Network engineering is shifting. Engineers are no longer just configuring devices — they’re defining how the network should behave.

This becomes even more critical as we move away from scripts and syntax. In an AI-driven world, our intent becomes the prompts that define the outcomes, and the systems will execute against them — and orient by leveraging the system of record
the structured context map for the infrastructure.

That's where the **NetBox Labs Platform** comes in, providing a system of record, along with discovery and validation, giving teams a complete picture of the network's intended and operational state.

Network engineering isn't going away — it's evolving.

The future isn't engineers writing better configurations. It's engineers defining better intent — and relying on systems that continuously enforce it.

TNS



Doug is a lifelong learner and educator, having focused his career on improving developer knowledge and experiences. A Google Developer Expert for the web, O'Reilly author, international keynote speaker, and a prolific blogger, he relishes in simplifying the complex. When...

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